

Public Economics

Lecture 5 · Budget Analysis and Deficit Financing

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Today's Questions

- Who actually decides what a government spends and taxes each year — and what stops that process from running an unlimited deficit?
- What does it even mean to “measure” a deficit correctly — nominal or real, cash or accrual, actual or cyclically adjusted?
- Do today's debts and deficits matter for the long run, and if so, why should we care?

Based on

Gruber, J. *Public Finance and Public Policy*, Ch. 4 (§4.1–4.5, Budget Analysis and Deficit Financing).

5.1 The Budget Process

5.2 Measuring the Budget's True Position

5.3 Do Debts and Deficits Matter? A Long-Run View

Deficits, Crowding Out, and Growth

5.1 The Budget Process

Three related concepts

- **Budget** = planned spending and revenue for a fiscal year.
- **Deficit** = spending $>$ revenue in a given year (a *flow*).
- **Debt** = accumulated past deficits (a *stock*).
- The EU's **Maastricht rules** cap the deficit at 3% of GDP and gross debt at 60% of GDP for member states
- Every question in this lecture is really about *how* we measure these numbers, and *why* the numbers matter beyond the headline percentage.

Austria's Budget Cycle

1. The **Bundesministerium für Finanzen (BMF)** drafts the annual budget law (*Bundesfinanzgesetz*, BFG) and the medium-term expenditure framework (*Bundesfinanzrahmengesetz*, BFRG).
2. The Finance Minister opens the process with the **Budgetrede** — the budget speech — in the Nationalrat, followed by a first reading.
3. The **Budgetausschuss** (budget committee) reviews both drafts in detail, supported by the **Budgetdienst** — Parliament's own independent budget office.
4. A structured, multi-day plenary debate follows: general debate, then a chapter-by-chapter (*Untergliederung*) detail debate, then the final vote — by **simple majority**; the Bundesrat has no blocking role.
5. After the fiscal year ends, the **Rechnungshof** (Court of Audit) prepares the final accounts (*Bundesrechnungsabschluss*) for Nationalrat approval.

- Austria's independent fiscal council traces back to the **Staatsschuldenausschuss** (1970); it was renamed and reconstituted as the **Fiskalrat** in **November 2013**.
- Institutionally housed at the **Oesterreichische Nationalbank (OeNB)** since 1997
- It is Austria's designated **Independent Fiscal Institution** under EU Regulation: it monitors compliance with national and EU fiscal rules and publishes an annual macro-fiscal outlook.

Keeping Bund, Länder, and Gemeinden in Line

- Austria is a federal state: the Maastricht deficit and debt ceilings apply to *general* government — federal, state (*Länder*), and local (*Gemeinden*) combined.
- The **Österreichischer Stabilitätspakt (ÖStP 2012)**, in force since January 2012, is the domestic mechanism that allocates deficit and debt targets — and sanctions for missing them — across these three levels.
- Unlike Germany's constitutionally entrenched *Schuldenbremse*, attempts to anchor one in Austria's federal constitution failed to reach the required two-thirds majority — first in **December 2011**, then again in the Bundesrat in **October 2019**.
- Austria's debt brake exists only as *ordinary law*
- The EU Council opened an **excessive deficit procedure** against Austria on 8 July 2025, triggered by a 2024 deficit of 4.7% of GDP — with an approved path requiring the deficit below 3% by 2028.

Application: What Happens When a Fiscal Rule Is Relaxed?

Grembi, Nannicini & Troiano (2016) study a natural experiment in fiscal-rule design: Italy's 1999 national rule restricted municipal deficit growth, but was **relaxed in 2001** for municipalities under 5,000 residents.

- Comparing municipalities just above and below the 5,000-resident cutoff, the study finds that once freed from the restraint, municipalities let **taxes fall** and **deficits rise**.
- The effect was **larger** where the mayor could be re-elected, where more parties competed, and where voters were older — consistent with a political-economy story, not just a mechanical budget-constraint effect.

Source: Grembi, Nannicini & Troiano, "Do Fiscal Rules Matter?" *AEJ: Applied Economics* 8(3), 2016.

5.2 Measuring the Budget's True Position

Real vs. Nominal: The Inflation Tax

- Inflation erodes the *real* value of outstanding government debt — effectively an “inflation tax” on whoever holds that debt.
- Approximation: inflation tax (% of GDP) $\approx$ debt-to-GDP ratio $\times$ inflation rate.

Austria, end-2024 debt-to-GDP	79.9%
Austria, average 2024 inflation (HICP)	2.9%
Implied inflation tax	$\approx$ 2.3% of GDP

A deficit measured in purely nominal terms overstates how much the government’s real debt burden actually grew.

Cyclical vs. Structural Deficits

- Deficits move **automatically** with the business cycle: recessions cut tax revenue and raise unemployment-related spending, even with no policy change (“automatic stabilizers”).
- To judge whether fiscal *policy* is loose or tight, economists strip out the cyclical component to get the **structural (cyclically adjusted) balance**.

Cash vs. Accrual Accounting: The International Experience

- **Cash accounting:** counts revenue/spending when money actually moves. Simple, but treats a euro spent on a durable investment (a bridge) the same as a euro spent on a one-off transfer.
- **Accrual (capital) accounting:** spreads the cost of long-lived investments over their useful life, like a firm's balance sheet — but is harder to implement and easier to manipulate through valuation assumptions.

International experience is mixed: **New Zealand** was the first country to fully implement accrual/capital budgeting (1989) and still uses it; **Denmark, Finland, and the Netherlands** adopted and later discontinued elements of it; **Britain** implemented it in 2001 and used it for its 2013 “Investing in Britain's Future” plan.

Austria's Own Shift: The Haushaltsrechtsreform

Austria itself moved toward accrual-style budgeting at the federal level, in two stages:

- **Stage 1 (2009):** introduced the multi-year BFRG framework and enabled genuine two-year budgets.
- **Stage 2 (2013):** a wholly new Bundeshaushaltsgesetz (BHG 2013) — full *Wirkungsorientierte Haushaltsführung* (outcome-oriented budgeting, including mandatory gender budgeting), replacing pure cash accounting (*Kameralistik*) with **accrual double-entry bookkeeping** across three linked statements:

Statement	What it tracks
Ergebnishaushalt	Operating result (like a P&L)
Finanzierungshaushalt	Cash flow (the traditional cash position)
Vermögensrechnung	Balance sheet of assets and liabilities

Static vs. Dynamic Scoring

- **Static scoring:** estimates a policy's budget effect holding the size of the economy fixed.
- **Dynamic scoring:** lets the estimate account for how the policy itself changes growth, employment, or behavior — and therefore future tax revenue.
- Dynamic estimates are usually *smaller* in absolute size than static ones for a tax cut, since some of the revenue loss is offset by higher growth — but they rest on more assumptions, and are correspondingly more contested.

5.3 Do Debts and Deficits Matter? A Long-Run View

The Tool: Present Discounted Value

Present discounted value (PDV)

A euro received (or saved) t years from now is worth less than a euro today. To compare payments across time, discount each future flow by the interest rate r :

$$PDV = \frac{F_1}{(1+r)} + \frac{F_2}{(1+r)^2} + \cdots + \frac{F_T}{(1+r)^T}$$

Any government decision that trades an upfront cost for a stream of future costs or savings — building infrastructure, cutting taxes today and raising them later, borrowing now to invest — has to be evaluated this way, **not** by comparing raw undiscounted totals.

Worked Example: Should the Government Pave the Road?

A government can pave a highway with a new “wear-proof” material. Paving costs €4 billion *today*, but saves €400 million in maintenance costs at the end of *each* of the next 10 years.

$$PDV(\text{savings}) = \sum_{t=1}^{10} \frac{\text{€}0.4\text{bn}}{(1+r)^t}$$

Discount rate r	0%	4%	8%
PDV of savings	€4.00bn	€3.24bn	€2.68bn
Worth it (cost = €4bn)?	break-even	No	No

Why Current Debt Labels Can Mislead

- Governments make huge **implicit** promises — future pension and healthcare benefits — that never appear in the officially reported “debt” figure, which counts only explicit borrowing.
- Relabeling a payment (e.g. calling a benefit cut a “tax increase,” or vice versa) can leave the government’s *true* long-run fiscal position completely unchanged, even though the headline deficit number moves.

This is why economists supplement the official, cash-based debt figure with a much broader concept: the **present value of all future obligations**, explicit and implicit alike.

Explicit vs. Implicit Debt: Austria's Sustainability Gap

- **Explicit debt:** bonded liabilities of federal, state, and local government — can only be reduced by repayment.
- **Implicit debt:** the present value of promises the state has made under current law but hasn't borrowed for — above all, pay-as-you-go pension commitments. Can only be reduced by reform.

	Austria	EU-28 average
Explicit debt (% GDP)	84%	85%
Implicit debt (% GDP)	177%	132%
Sustainability gap	260%	217%

Source: European Commission/Eurostat; Forschungszentrum Generationenverträge (2017).

Measuring Long-Run Sustainability: Austria's Pension System

The European Commission's **2024 Ageing Report** projects Austria's public pension expenditure from 2022 to 2070:

	2022	Peak (≈ 2032)	2070
Old-age & early pensions (% GDP)	11.4%	12.9%	—
Total public pension spending (% GDP)	13.7%	—	14.0%

- Spending rises as the baby-boomer generation retires (peaking around 2032), then eases back down — a net increase of only **~ 0.3 percentage points of GDP** over the full 48-year horizon.
- Domestically, the **Alterssicherungskommission** produces an annual 5-year assessment (due by 30 November) and a long-term report every three years, currently projecting to 2050.

How Far Ahead Should We Look? Budget Windows

- A short budget window (one year) misses obligations that only bite later; a very long one (75 years, as US long-run scores sometimes use) is dominated by uncertain assumptions about growth, interest rates, and demographics.
- Austria has a rolling **4-year** window (§5.1) — extendable to **7 years** under the EU's reformed fiscal framework if reforms and investments are committed to.
- There is no “correct” window — only a trade-off between capturing genuinely long-run obligations and trusting projections far enough out that the numbers stop being credible.

Deficits, Crowding Out, and Growth

The OLG Model: A Microfoundation for Crowding Out

- The **overlapping-generations (OLG)** model gives the crowding-out story a few slides ahead an explicit microfoundation: each generation lives two periods, working when young and living off savings when old.
- Young-period budget: $c_t^y + s_t = w_t$. Old-period budget: $c_t^o = (1 + r_{t+1})s_t$.
- Combined into one intertemporal budget constraint:

$$c_t^y + \frac{c_t^o}{1 + r_{t+1}} = w_t$$

- Households choose c_t^y, c_t^o — and thus savings s_t — to maximize lifetime utility subject to this constraint.

Government in the OLG Model

- Public expenditures are normalized to zero. Every member of the **working** population pays a lump-sum tax τ_t ; the **retired** pay no tax.
- Generation t 's intertemporal budget constraint becomes:

$$c_t^y + \frac{c_t^o}{1 + r_{t+1}} = w_t - \tau_t$$

- Baseline, no debt:** the government simply balances its budget every period, taxing exactly what it spends.

The Policy Experiment: A Deficit-Financed Tax Cut

Suppose in period t the government raises per-capita debt by Δ and cuts this generation's tax by the same amount:

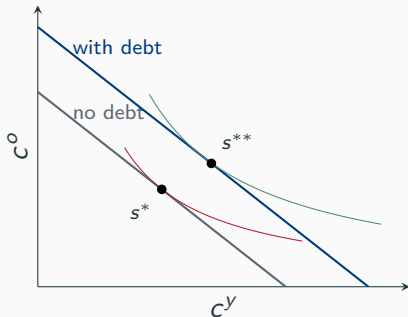
$$\tau_t + \Delta = 0$$

The debt, plus interest, must be repaid the following period — by this *same* generation, now old:

$$\tau_{t+1} - (1 + r_{t+1})\Delta = 0 \quad \Longleftrightarrow \quad \tau_{t+1} = (1 + r_{t+1})\Delta$$

So the experiment asks: what happens if one generation's *own* lifetime tax burden is shifted from young age to old age, bridged by government debt in between?

The New Budget Constraint



Substituting $\tau_t = -\Delta$ into the constraint:

$$w_t + \Delta = c_t^y + \frac{c_t^o}{1 + r_{t+1}}$$

- The budget line shifts out in parallel by Δ .
- If c^y, c^o are normal goods, the household consumes more in **both** periods.
- c^y rises by $0 < dc^y < \Delta$ — so saving $s_t = w_t + \Delta - c_t^y$ rises, but by **less than** Δ .

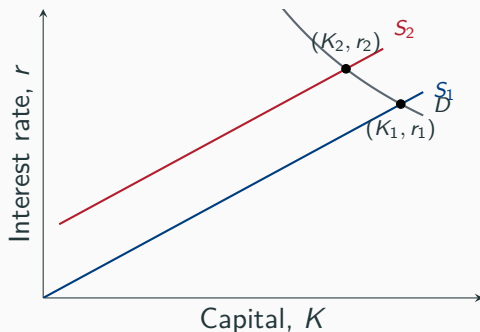
Effects on the Capital Market

- Public saving falls by the full Δ ; private saving rises by less than Δ — so **total** national saving falls.
- In the capital market (next slide): the savings curve S shifts left, the interest rate r rises, and equilibrium capital K falls.
- We might call this **crowding out** of private investment by public demand for credit — it reduces the capital stock, which lowers the income of the population in the long run.

Normative Questions: Redistribution and Welfare

- The OLG model shows that public debt raises interest rates and lowers the capital stock in the long run.
- **Future generations** are worse off (lower capital $\Rightarrow$ lower wages); the **current generation** is clearly better off (it pays less tax).
- Assessing the overall welfare effect requires taking a stance on how to weight the welfare of different generations.
- If technological progress raises future incomes anyway, future generations may end up better off than the current one *regardless* — which some use to argue for redistribution from future to current generations. This is ultimately a normative judgment, not something economics alone can settle.

How a Deficit Can Crowd Out Private Investment



- Firms' demand for capital, D , meets the supply of private savings — a deficit means the state borrows from that same pool, which would otherwise fund private investment.
- This shifts the supply of loanable funds available to private borrowers left, from S_1 to S_2 .
- Equilibrium moves from (K_1, r_1) to (K_2, r_2) : interest rates **rise**, private capital investment **falls**.
- This is the standard **crowding-out** mechanism.

Ricardian Equivalence

- Ricardo (1817) posed the question, Barro (1974) gave it a modern answer: if households are forward-looking and leave positive bequests, a debt-financed tax cut today is simply saved — to cover the higher taxes needed to repay the debt later.
- Under Ricardian equivalence, financing government spending through taxes or through debt is **completely equivalent**: consumption, saving, investment, and interest rates are all unaffected.
- The result needs strong conditions: positive bequests, no differences in family size across households, non-distortionary taxes, and perfect capital markets.

Empirical evidence favors the crowding-out story instead: deficits are associated with higher interest rates and lower investment (e.g. Ardagna et al. 2007; Gale & Orszag 2003).

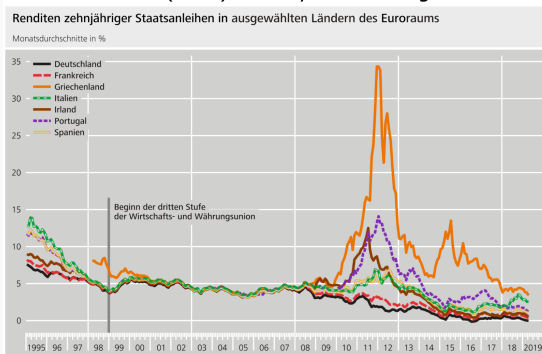
Open Capital Markets Blunt Crowding Out

- The crowding-out story assumes savings are supplied *domestically* and inelastically. With fully open, integrated international capital markets, a country can borrow from the whole world's savings pool — so its own deficit barely moves the interest rate it faces.
- As one country inside the euro area's single capital market, Austria effectively faces a supply of savings that is close to perfectly elastic — far more so than a large economy like the US, where domestic savings still matter enough that some crowding-out concern remains empirically credible.

Sovereign Debt Crises: When Expectations Turn

- Reinhart & Rogoff (2009, *This Time Is Different*): once creditors doubt a government can service its debt, they demand higher interest rates and sell existing bonds.
- Higher rates make the debt harder to service, which can **confirm** the very default fear that triggered them — a self-fulfilling crisis.
- A state cannot be liquidated: an unsustainable debt load ends in negotiated restructuring — a **haircut** for creditors.

Evidence: The Eurozone Debt Crisis



Ten-year sovereign yields for Germany, France, Greece, Italy, Ireland, Portugal, and Spain, 1995–2019: a single currency didn't stop borrowing costs from diverging sharply once markets began pricing in default risk.

Source: Deutsche Bundesbank.

The Secular Stagnation Puzzle

Real interest rates across the industrialized world have fallen sharply and persistently over the past four decades

Furman & Summers (2020) show this decline holds across Canada, France, Germany, Italy, Japan, the UK, and the US alike — a genuinely worldwide phenomenon, not a US-specific one.

Period	Real interest rate
Mid-to-late 1980s, average across industrialized nations	>5%
January 2000 (US)	4.3%
Pre-Covid, February 2020 (US)	−0.1%

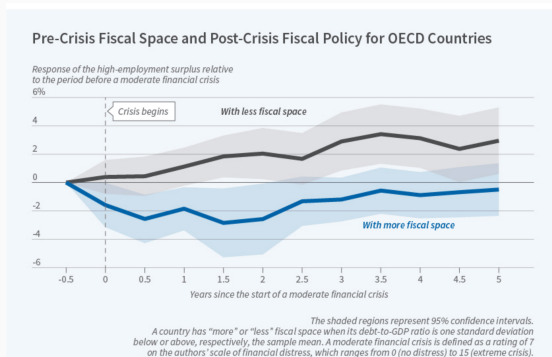
Austria and the Negative-Rate Era

- The ECB cut its deposit facility rate below zero in **June 2014** (to -0.10%), deepening it to -0.50% by **September 2019** — and rates stayed negative until **July 2022**.
- Through most of that period, Austrian (and German) long-term government bond yields traded at or below zero: investors were, in effect, *paying* the government to hold their savings.
- This an example of secular-stagnation puzzle

Can Public Debt Be Justified?

- **Tax smoothing** (Barro, 1979): the deadweight loss of a tax rises more than proportionally with the tax rate, so smoothing the rate over time — borrowing in bad years, repaying in good ones — minimizes the total distortion versus adjusting taxes every year.
- **Anti-cyclical stabilization**: debt lets governments decouple spending from business-cycle tax revenue — but this only justifies *temporary* debt, not a decades-long upward trend.
- **Musgrave's pay-as-you-use principle**: generations who benefit from public investment (infrastructure, education) should share its cost — but this only justifies debt when net public investment is positive, which it often has not been.

Evidence: Fiscal Space Pays Off in a Crisis



OECD countries that entered a financial crisis with **more** fiscal space (lower debt) ran larger deficits in response — and recovered faster.

Source: Romer & Romer (2019), NBER WP 25768.

The “New View” of Deficits

- If persistently low (or negative) real interest rates mean deficits no longer visibly crowd out private investment, some economists argue fiscal sustainability should be judged by the **ratio of interest payments to GDP**, not the debt-to-GDP ratio itself.
- This does **not** mean deficits are “free”: debts must still eventually be serviced, and the argument only strengthens the case for spending that pays for itself — productive investment with a return above the (low) cost of borrowing.

Thank you!

Questions and discussion